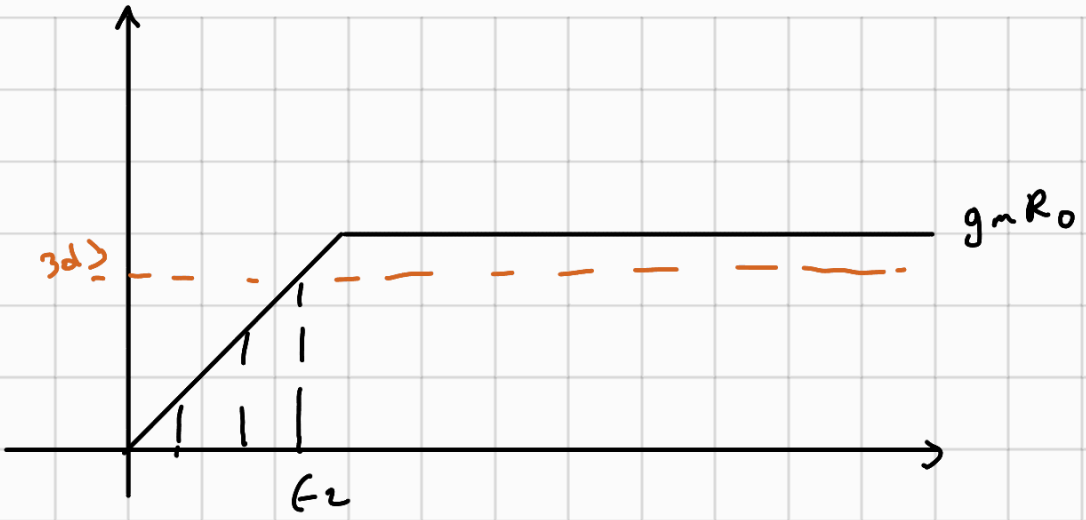
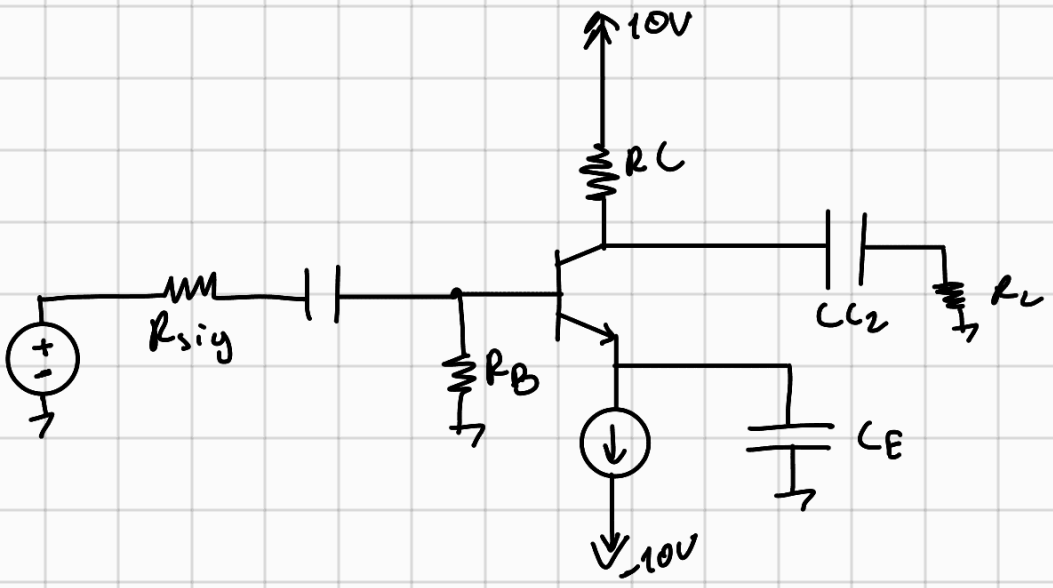


$\tau_1 = R_1 C_1$ → f_2 کوچک‌تر

$\tau_2 = R_2 C_2$ کوچک‌تر

$\tau_3 = R_3 C_3$

$\frac{1}{2\pi R_C C}$ ↓ ↑



Calculate the time constants for all poles at low frequency and estimate f_L. Calculate A_m = ?

$I = 1mA$ $\beta = 100$

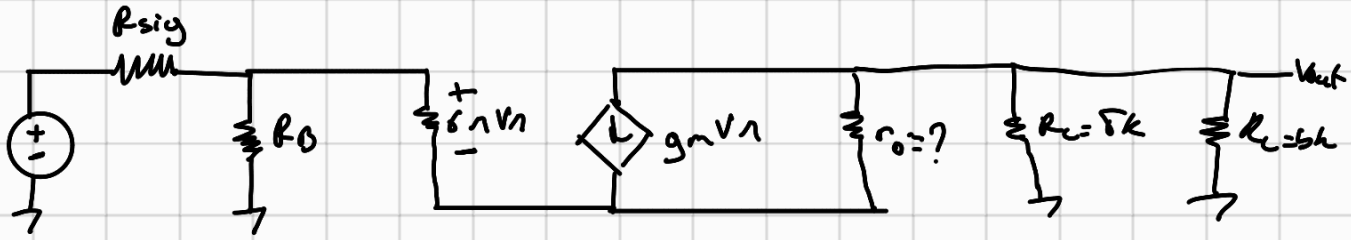
$R_B = 100k\Omega$ $V_A = 100V$

$R_C = 8k\Omega$ $C_{C1} = C_{C2} = C_E = 1\mu F$

$R_{sig} = 5k\Omega$

$R_L = 5k\Omega$

for mid band gain



$$g_m = \frac{I_e}{V_t} = \frac{1\,mA}{25\,mA} = 40\,mS$$

$$r_e = 1/g_m \cong 25\,\Omega$$

$$r_\pi = 25\,\Omega \times 101 = 2{,}5\,k\Omega$$

$$r_o = \frac{V_A}{I_c} = \frac{100}{1} = 100\,k\Omega$$

$$\frac{v_{out}}{v_{sig}} = \frac{v_{out}}{v_\pi} \cdot \frac{v_\pi}{v_{sig}}$$

$$v_{out} = -g_m v_\pi (r_o \,//\, R_c \,//\, R_L)$$

$$\frac{v_{out}}{v_\pi} = -40(2{,}81)$$

$$\frac{v_{out}}{v_\pi} = -116{,}5$$

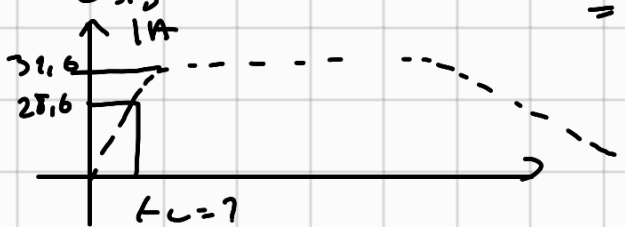
$$v_\pi = v_{sig} \frac{(R_B // r_\pi)}{R_{sig} + (R_B // r_\pi)}$$

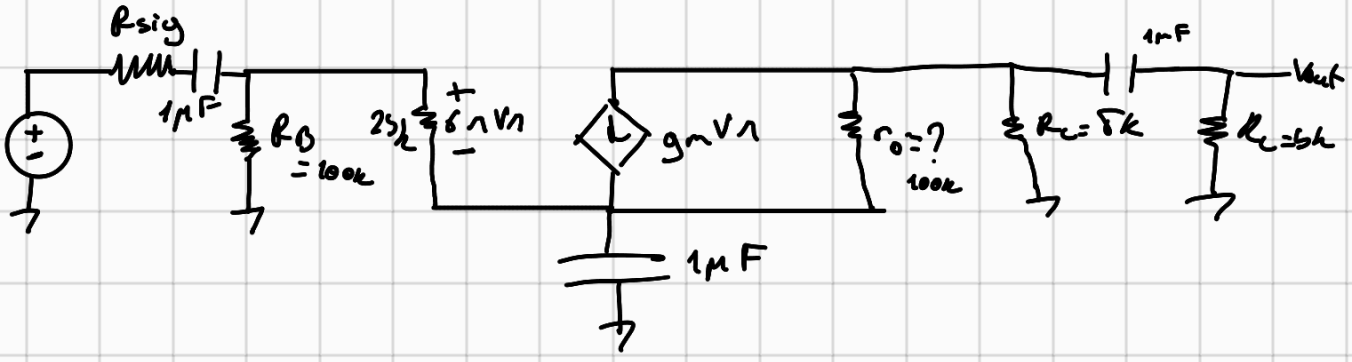
$$= \frac{2{,}43}{7{,}43} v_{sig}$$

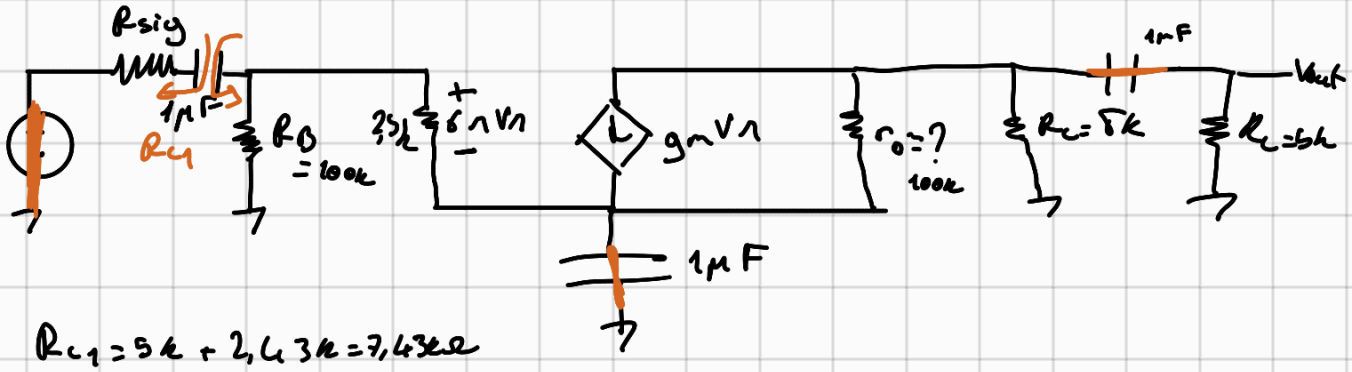
$$\frac{v_\pi}{v_{sig}} = \frac{2{,}43}{7{,}43} = 0{,}32$$

$$\frac{v_{out}}{v_{sig}} = -116{,}5 \times 0{,}32 = -38{,}1\ [V/V]$$

$$= 31{,}6\ dB$$



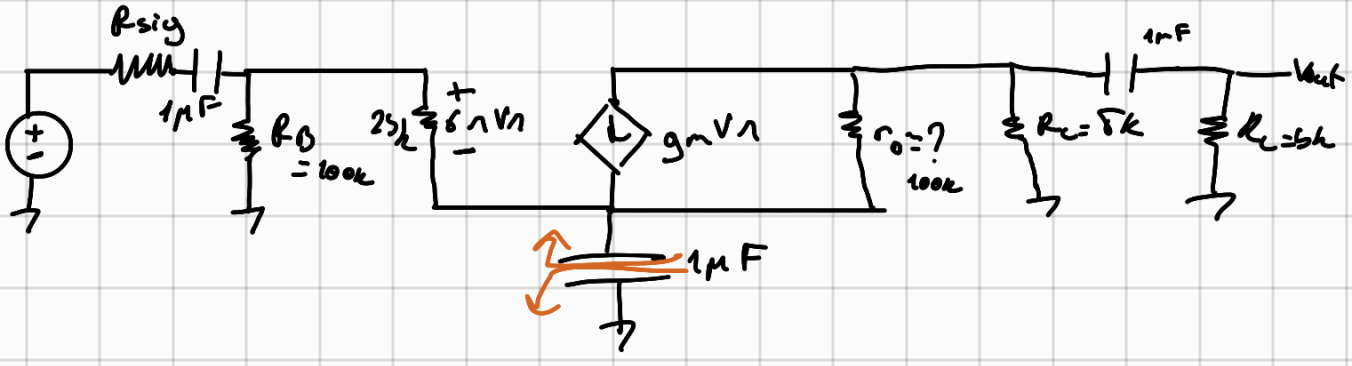




$R_{c1} = 5k + 2,43k = 7,43k\Omega$

$z_1 = R_{c1} C_{c1} = 7,43k\Omega \cdot 1\mu F$

$z_1 = 7,43\ msec$



$R_E = \frac{2,5k + 4,7\Omega}{101} \simeq 72$

$z_2 = R_E C_E = 72\Omega \times 1\mu F = 72\mu s$

$z_3 = R_{c2} C_{c2}$

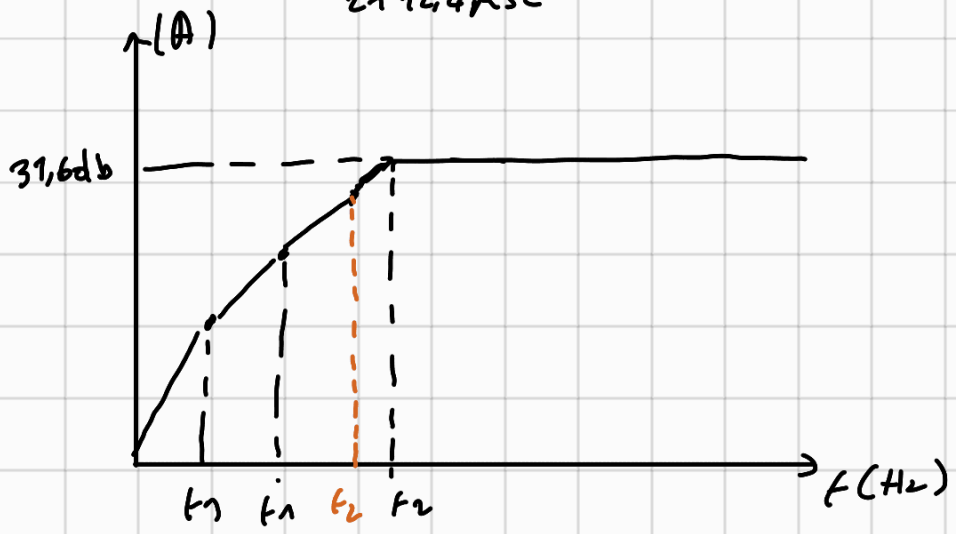
$R_{c2} = 5k + 7,4 = 12,4k\Omega$

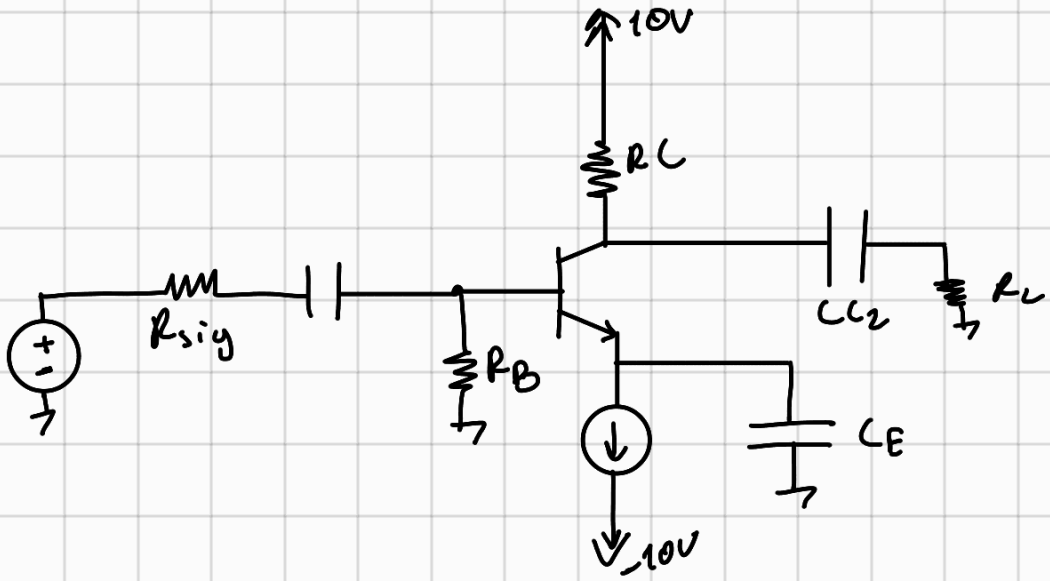
$z_3 = 12,4\ k\Omega \times 1\mu F = 12,4\ msec$

$f_1 = \frac{1}{2\pi z_1} = \frac{1}{2\pi \cdot 7,43} \simeq 21\ Hz$

$f_2 = \frac{1}{2\pi z_2} = \frac{1}{2\pi \cdot 72\ nsec} \simeq 2,2\ kHz$

$f_1 = \frac{1}{2\pi\tau_3} = \frac{1}{2\pi\, 12{,}4\,msec} \cong 12{,}8\,Hz$





Calculate the time constants for all poles at high frequency and estimate f_L. Calculate A_m = ?

$I = 1mA$ $\quad \beta = 100$

$R_B = 100k\Omega$ $\quad V_A = 100V$

$R_C = 8k\Omega$ $\quad C_{C1} = C_{C2} = C_E = 1\mu F$

$R_{sig} = 5k\Omega$ $\quad f_t = 800\,MHz$

$R_L = 5k\Omega$ $\quad r_x = 50\Omega$

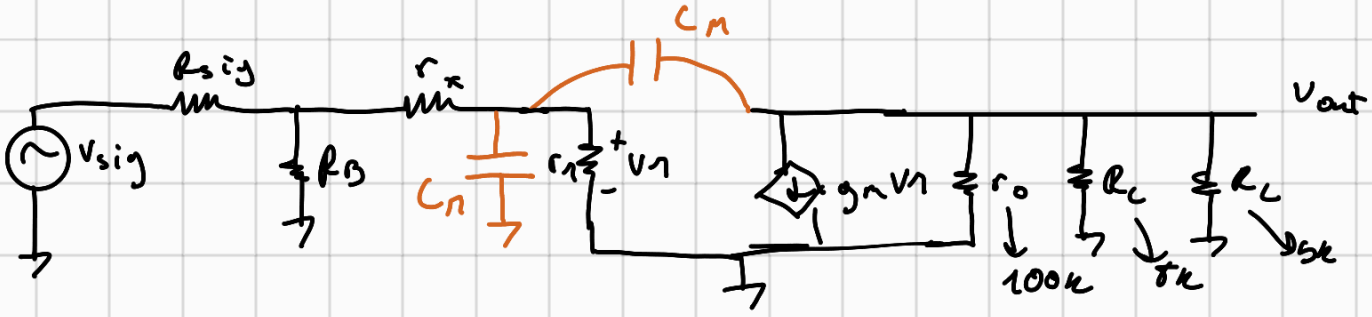
$C_\mu = 1pF$

$g_m = \frac{I_C}{V_t}$

$$f_t = \frac{g_m}{2\pi(C_\mu + C_x)} = \frac{40mS}{2\pi(1pF + C_x)} = 700 \cdot 10^6$$

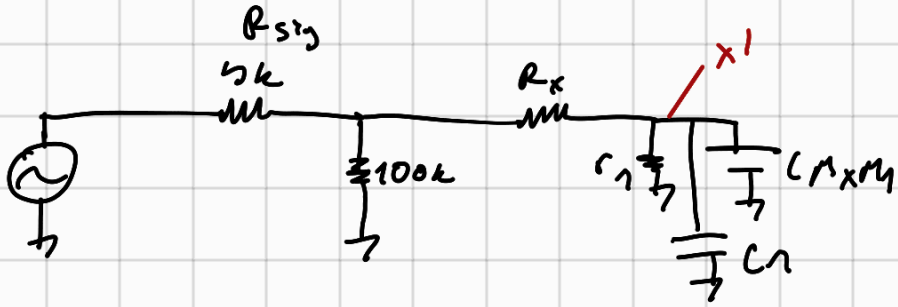
$$\frac{40 \cdot 10^{-3}}{} = 1600\Omega \cdot 10^{-6} + 1600\Omega \cdot 10^{6} C_x$$

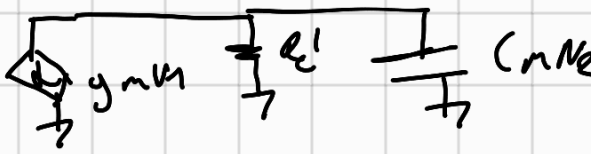
$$C_x = 6{,}99\,pF$$

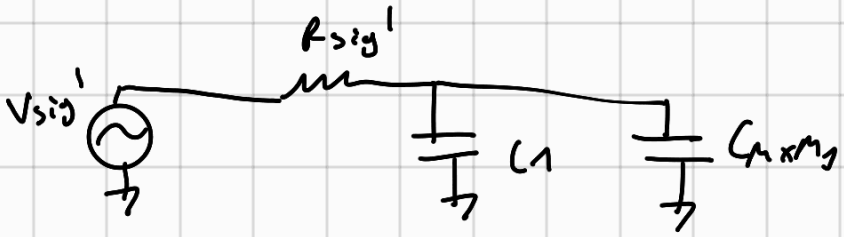


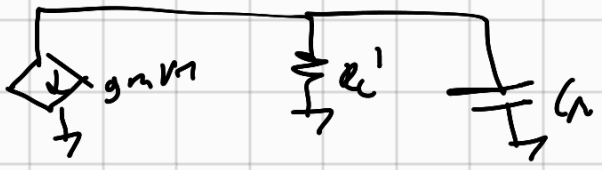
$$f_t = \frac{g_m}{2\pi(C_\mu + C_\pi)}$$

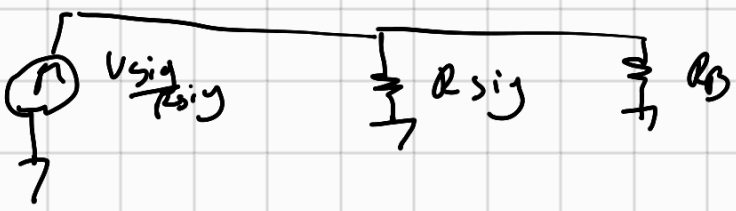
$$\Rightarrow C_x = 6{,}99\,p$$





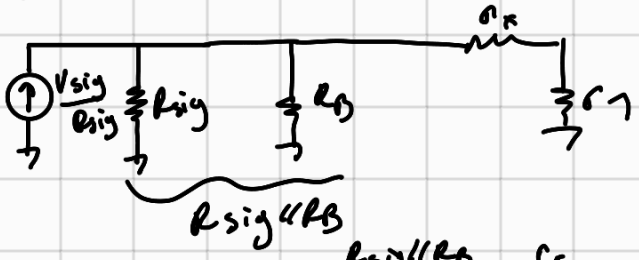


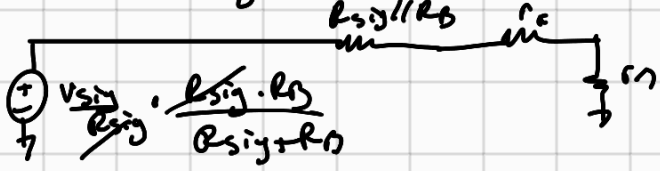




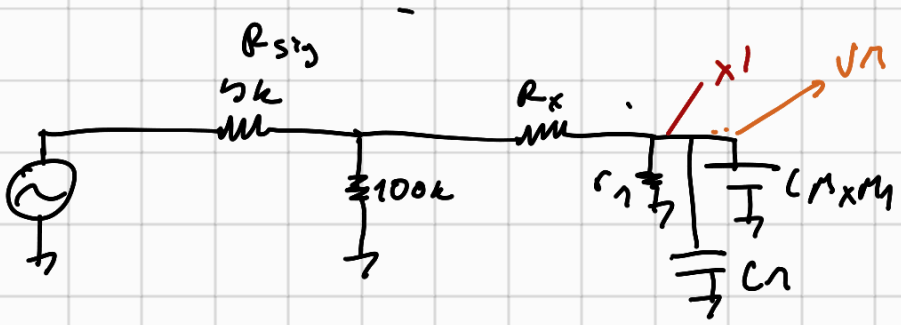


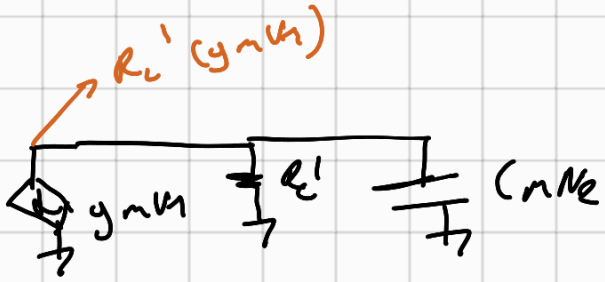
Vsig' kaynak dönüşümü gaparakbulalı





$$V = V_{sig} \cdot \frac{R_B}{R_{sig}+R_B} \cdot \frac{1}{(R_{sig}//R_B + r_x + r_\pi)} \cdot r_\pi$$





$$K = g_m R_L'$$

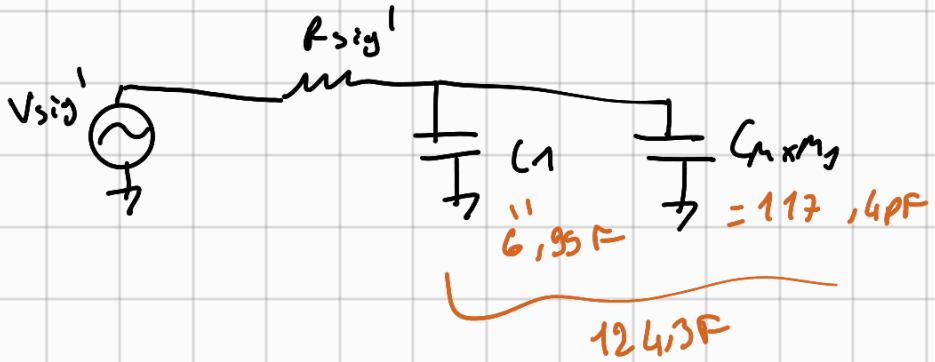
$$Z = \frac{1}{j\omega C}$$

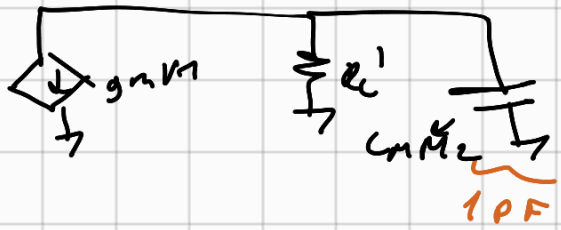
$$M_1 = (1 + g_m R_L') = 117{,}4$$

$$M_2 \left(1 + \frac{1}{g_m R_L'}\right) = 1$$

($g_m R_L'$: « beaucoup »)

$$R_L' = 100k \,//\, 8k \,//\, 5k\Omega = 2{,}91k\Omega$$



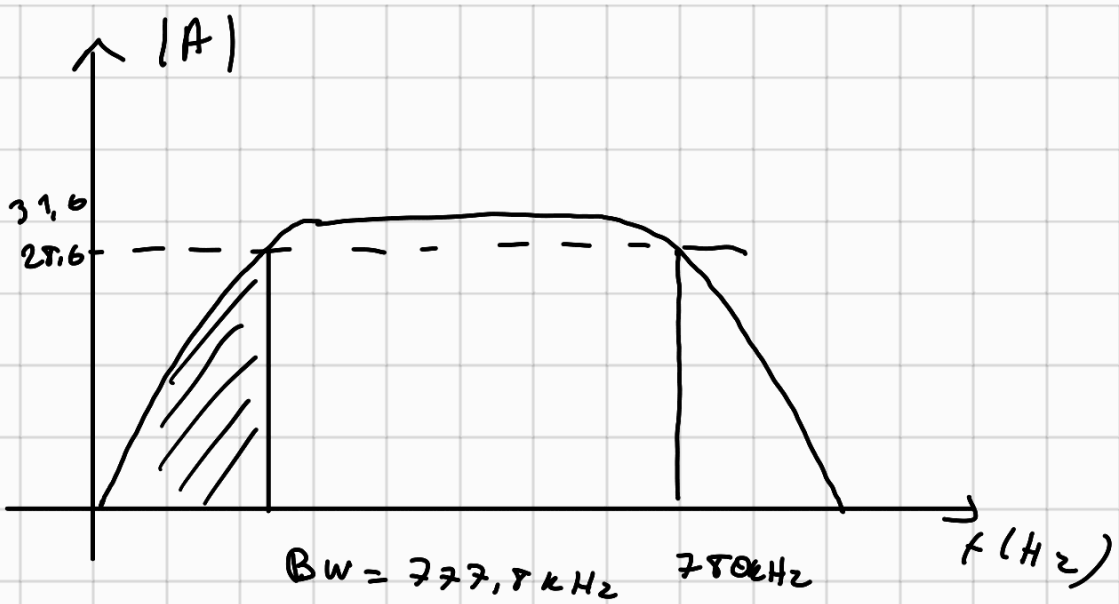


$$R_{sig}' = 2{,}5k \,//\, (4{,}76k)$$

$$= 1{,}64k\Omega$$

$$f_1 = \frac{1}{2\pi \cdot 1{,}64 \cdot 10^3 \cdot 124{,}3 \cdot 10^{-12}} = 780kHz$$

$$f_2 = \frac{1}{2\pi (2{,}91k \cdot 1 \cdot 10^{-12}} \approx 55MHz$$


|A|
31,6
25,6
f (Hz)
BW = 777,8 kHz
750kHz